

CS60092: INFORMATION RETRIEVAL

Programming Assignment 2 Vector Space Model and Ranked Retrieval

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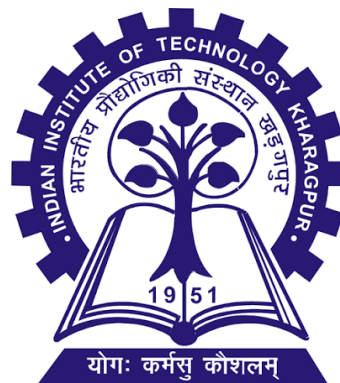
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1. Problem Statement

The objective of this work is to develop and evaluate a ranked Information Retrieval (IR) system for the Cranfield collection using an open-source retrieval framework such as PyTerrier or Apache Lucene.

Let the document collection and query set be

$$\mathcal{D} = \{d_1, d_2, \dots, d_N\}, \quad \mathcal{Q} = \{q_1, q_2, \dots, q_M\}. \quad (1)$$

Given a query q_i , a retrieval model M with parameter configuration θ assigns a score

$$S(q_i, d_j \mid M, \theta) \quad (2)$$

to each document d_j , producing a ranked list

$$R_{M,\theta}(q_i) = (d_{i_1}, d_{i_2}, \dots, d_{i_N}), \quad (3)$$

ordered by decreasing retrieval score.

The ranked results are evaluated against the Cranfield relevance judgments J using standard retrieval effectiveness measures, including Mean Average Precision (MAP), Mean Reciprocal Rank (MRR), Precision at k ($P@k$), and Normalized Discounted Cumulative Gain ($nDCG@k$). The effectiveness of a retrieval configuration is defined as

$$E(M, \theta) = \mathcal{E}(R_{M,\theta}(\mathcal{Q}), J). \quad (4)$$

Multiple sparse retrieval models and their parameter settings are investigated. The optimal configuration is defined by

$$(M^*, \theta^*) = \arg \max_{M \in \mathcal{M}, \theta \in \Theta_M} E(M, \theta). \quad (5)$$

The study therefore focuses on comparing sparse retrieval models, optimising their parameters, and identifying the configuration that achieves the best retrieval effectiveness on the Cranfield test queries.

2. Implementation Details

The retrieval system was implemented using **PyTerrier**. The overall implementation consisted of dataset preparation, preprocessing, indexing, retrieval, and evaluation.

2.1 Dataset and Preprocessing

The experiments were performed on the **Cranfield collection**, consisting of documents, queries, and corresponding relevance judgments (qrels). The documents and queries were processed using the standard preprocessing pipeline, including tokenization and term processing. The same preprocessing procedure was used throughout all experiments.

2.2 Indexing

The preprocessed documents were indexed using the **Terrier inverted-indexing mechanism**. The same index was reused for the different retrieval models to maintain a consistent experimental setup. For field-weighted retrieval, title and document text were indexed as separate fields.

2.3 Retrieval

The indexed collection was searched using different sparse retrieval models, including **TF**, **TF-IDF**, and **Lemur TF-IDF**. Parameter tuning was performed using **grid search**. **Field-weighted TF-IDF** was used to combine title and text scores, while **RM3 pseudo-relevance feedback** was applied for query expansion.

2.4 Evaluation

All retrieval configurations were evaluated against the Cranfield qrels using **MAP**, **MRR**, **P@5**, **P@10**, and **nDCG@10**. MAP was used as the primary measure for model selection. Indexing time and search time were also recorded to compare computational efficiency.

3. Plan of experiments

The retrieval system was developed through a progressive sequence of experiments on the Cranfield collection. The same document collection, preprocessing procedure, query set, relevance judg-

ments, and evaluation metrics were maintained throughout the experiments to ensure a consistent comparison. The experiments focused exclusively on sparse, word-based retrieval methods.

3.1 Establishing and Validating the TF-IDF Baseline

The first experiment implemented a conventional TF-IDF retrieval model. For a term t and document d , the TF-IDF representation can be expressed as

$$\text{tf-idf}_{t,d} = \text{tf}_{t,d} \log \left(\frac{N}{\text{df}_t} \right). \quad (6)$$

The initial evaluation produced unexpectedly low scores:

$$\text{MAP} = 0.00871, \quad \text{MRR} = 0.02068. \quad (7)$$

Rather than interpreting this as a failure of TF-IDF, the evaluation pipeline was investigated. The document and query data types were verified, and manual inspection of retrieved documents showed that TF-IDF was capable of retrieving relevant documents.

The issue was subsequently identified as a query-ID alignment problem. The identifiers appearing after the `.I` field in `cran.qry` were not aligned with the sequential query identifiers used by the `qrels`. Therefore, queries were reassigned according to their order in the query file:

$$q_i \longrightarrow \text{qid} = i, \quad i = 1, \dots, 225. \quad (8)$$

After correcting the alignment, the TF-IDF results increased substantially, as shown in Table 1.

Table 1: Baseline TF-IDF retrieval performance

Model	MAP	MRR	P@5	P@10	nDCG@10	Index Time (s)	Search Time (s)
TF-IDF	0.318897	0.554530	0.328889	0.236444	0.346963	4.632597	7.051320

This corrected TF-IDF system was therefore adopted as the principal baseline.

Following the establishment of the corrected baseline, a parameter fine-tuning experiment was conducted to determine whether the TF-IDF configuration could be further improved. A grid search was performed over the available `c` parameter values, with each configuration evaluated using the same Cranfield queries and relevance judgments. The results showed that the **default value of `c`** achieved the strongest overall retrieval effectiveness among the configurations tested. In particular, no alternative value of `c` produced a consistent improvement over the default configuration across

the evaluation measures. Therefore, the default TF-IDF configuration was retained as the optimal TF-IDF setting and used in subsequent experiments.

It is important to distinguish retrieval scores from relevance judgments. The retrieval function produces continuous scores used for ranking, whereas the qrels provide ground-truth relevance labels. Consequently, the two quantities are not expected to have the same numerical values.

3.2 Comparison of Sparse Retrieval Models

After establishing a valid baseline, three sparse term-weighting models were compared under identical conditions: TF, TF-IDF, Lemur TF-IDF. The comparison was intended to determine whether alternative sparse weighting schemes could improve upon the standard TF-IDF representation. The results of this comparison are presented in Table 2

Table 2: Comparison of sparse retrieval models

Model	MAP	MRR	P@5	P@10	nDCG@10
TF	0.1934	0.4297	0.1947	0.1516	0.2235
TF-IDF	0.3189	0.5545	0.3289	0.2364	0.3470
Lemur TF-IDF	0.2945	0.5129	0.2978	0.2276	0.3235

TF-IDF produced the strongest overall performance at this stage. Other models such as BM25, PL2, and DLM were also explored during preliminary experimentation, but they were not retained for the final comparison since the final system was required to remain within the sparse vector-space retrieval setting.

3.3 Lemur TF-IDF Enhancement

Following the initial model comparison, the Lemur TF-IDF model was further investigated using parameter tuning and two sparse vector-space improvements: Rocchio pseudo-relevance feedback and title-term boosting.

First, a grid search over k_1 and b was performed, with the best configuration identified as $k_1 = 2.0$ and $b = 0.25$. This tuned configuration was used as the basis for all subsequent experiments.

Rocchio pseudo-relevance feedback was then applied to the tuned Lemur TF-IDF ranking. The top-ranked documents were treated as pseudo-relevant, and the query was expanded using their TF-IDF centroid. The number of feedback documents and the feedback weight β were tuned over

$$\text{num_fb_docs} \in \{3, 5, 10, 15, 20\}, \quad \beta \in \{0.25, 0.5, 0.75, 1.0\}. \quad (9)$$

To control noisy query expansion, a further experiment retained only the top- K highest-weight feedback terms, with

$$K \in \{5, 10, 20, 40, 80\}, \quad (10)$$

along with an unpruned configuration.

Finally, title-term boosting was investigated by repeating preprocessed title terms before indexing. Boost factors of

$$B \in \{1, 2, 3, 5, 8\} \quad (11)$$

were evaluated while keeping the optimized Lemur TF-IDF parameters fixed.

These experiments were designed to assess whether parameter optimization, pseudo-relevance feedback, selective query expansion, and title weighting could improve the tuned Lemur TF-IDF model. The final comparison of these configurations is presented separately in the subsequent results section.

3.4 Weighted TF-IDF Retrieval

To further exploit the document structure, a weighted TF-IDF model was evaluated by retrieving independently from the title and text fields. Since the two fields can produce scores on different scales, the retrieval scores were first normalized independently for each query using Min–Max normalization:

$$S_{\text{norm}}(d, q) = \frac{S(d, q) - S_{\min}(q)}{S_{\max}(q) - S_{\min}(q)}. \quad (12)$$

The normalized title and text scores were then combined using a weighted sum:

$$S_{\text{weighted}}(d, q) = w_T S_T(d, q) + w_W S_W(d, q), \quad (13)$$

where w_T and w_W denote the title and text weights, respectively, with $w_T + w_W = 1$.

The title weight was initially varied from 0 to 1 in increments of 0.1. A finer search was subsequently performed between 0.10 and 0.30 to identify the best-performing region. The configuration with the highest MAP was selected.

The best configuration assigned

$$w_T = 0.27, \quad w_W = 0.73. \quad (14)$$

Its performance was compared with the established TF-IDF baseline, as shown in Table 3.

Table 3: Comparison of TF-IDF and weighted TF-IDF

Model	MAP	MRR	P@5	P@10	nDCG@10
TF-IDF	0.318897	0.554530	0.328889	0.236444	0.346963
Weighted TF-IDF	0.322927	0.577265	0.328889	0.245333	0.355195

The weighted model produced a modest improvement over the standard TF-IDF baseline across MAP, MRR, P@10, and nDCG@10, while P@5 remained unchanged. The selected weights indicate that the text field provided the larger contribution to retrieval effectiveness, while the title supplied additional useful evidence.

3.5 RM3 Pseudo-Relevance Feedback with TF-IDF

The TF-IDF retrieval model was further extended using RM3 pseudo-relevance feedback (PRF). RM3 uses the top-ranked documents from the initial retrieval to identify expansion terms and construct an expanded query, without using the ground-truth relevance judgments during retrieval.

Two parameters were tuned: the number of feedback documents (`fb_docs`) and the number of expansion terms (`fb_terms`). The initial parameter search identified

$$\text{fb_docs} = 30, \quad \text{fb_terms} = 20 \quad (15)$$

as the best configuration according to MAP on the full Cranfield evaluation set.

To further assess the selected parameters, additional experiments were performed using individual test queries. These experiments indicated that retaining 30 feedback documents while reducing the number of expansion terms to 15 produced better retrieval behaviour. Therefore, the final RM3 configuration adopted for the subsequent experiments was

$$\boxed{\text{fb_docs} = 30, \quad \text{fb_terms} = 15} \quad (16)$$

Thus, the value of 30 feedback documents was retained from the initial parameter search, while the number of expansion terms was reduced from 20 to 15 based on additional query-level validation. The same preprocessing and query-qrel alignment used in the preceding experiments were retained.

The combination of weighted TF-IDF with RM3 was also investigated. However, it did not provide a consistent improvement over standard TF-IDF with RM3 and was therefore not retained. A possible explanation is that weighted TF-IDF already modifies the ranking by combining separately normalized title and text scores, while RM3 derives expansion terms from the feedback-document distribution. Combining the two may therefore alter the feedback distribution and introduce less useful or redundant expansion terms. Consequently, the standard TF-IDF + RM3 configuration was selected for the final retrieval system.

3.6 Exploratory Adaptive Rocchio Pipeline

As an additional experiment, an adaptive retrieval pipeline combining RM3 pseudo-relevance feedback with a subsequent Rocchio feedback step was investigated. This experiment was conducted to explore the potential for further improvement beyond the selected TF-IDF + RM3 system.

The pipeline consisted of an initial TF-IDF retrieval, followed by two fixed RM3 configurations. The resulting top-ranked documents were then used for a Rocchio feedback step. The top five documents were treated as pseudo-relevant documents and incorporated into the query representation using a fixed feedback weight of $\beta = 20$. Queries containing at most eight processed terms were assigned to a robust RM3 branch, while longer queries used the best-MAP branch.

The complete procedure can be summarised as

$$q \rightarrow \text{TF-IDF} \rightarrow \text{RM3} \rightarrow \text{Rocchio} \rightarrow \text{Final Ranking}. \quad (17)$$

This experiment used a modified preprocessing and vector construction procedure compared with the main experiments. Since the experimental setup requires the preprocessing configuration to remain fixed, the adaptive Rocchio results are not considered directly comparable with the primary results and are not used for final model selection. Instead, this experiment was performed only to investigate the potential scope for further improvement within sparse retrieval.

No qrel information was used to modify individual queries during this procedure. The relevance judgments were used only for evaluation.

3.7 Final Retrieval Configuration

Based on the main experimental sequence, the final retrieval system was selected as TF-IDF with RM3 pseudo-relevance feedback. The baseline evaluation, parameter tuning, weighted TF-IDF experiment, and RM3 experiments showed that RM3 provided a useful improvement while

retaining the required sparse, word-based retrieval framework.

The final retrieval process is therefore

$$q \rightarrow \text{TF-IDF} \rightarrow \text{RM3} \rightarrow \text{Expanded Query} \rightarrow \text{TF-IDF} \quad (18)$$

The final RM3 configuration used 30 feedback documents and 15 expansion terms. The adaptive Rocchio experiment was treated only as an exploratory extension because it used a different preprocessing configuration. Consequently, the final reported system remains TF-IDF + RM3 under the established preprocessing and evaluation setup.

4. Results

The retrieval experiments were evaluated using five standard effectiveness measures: Mean Average Precision (MAP), Mean Reciprocal Rank (MRR), Precision at 5 (P@5), Precision at 10 (P@10), and Normalised Discounted Cumulative Gain at 10 (nDCG@10). Search time and indexing time was also recorded for the retrieval experiments. The experiments were conducted progressively, beginning with a comparison of the initial retrieval models and followed by parameter tuning, field weighting, and pseudo-relevance feedback. The results from each stage are presented below in Table 4.

Table 4: Overall comparison of the best-performing retrieval configurations

Configuration	MAP	MRR	P@5	P@10	nDCG@10
TF	0.193439	0.429683	0.194667	0.151556	0.223466
TF-IDF Baseline	0.318897	0.554530	0.328889	0.236444	0.346963
Lemur TF-IDF	0.301619	0.518089	0.304889	0.229778	0.332116
Lemur TF-IDF + Title Boost	0.308062	0.535583	0.314667	0.236889	0.339157
Weighted TF-IDF	0.322927	0.577265	0.328889	0.245333	0.355195
Lemur TF-IDF + Pseudo-Rocchio	0.340437	0.527942	0.359111	0.270667	0.366833
Lemur TF-IDF + Pseudo-Rocchio + Title Boost	0.344589	0.556455	0.351111	0.268444	0.374689
TF-IDF + RM3	0.353066	0.565110	0.360000	0.273778	0.384341

The overall results demonstrate a progressive improvement in retrieval effectiveness as increasingly sophisticated retrieval techniques were introduced.

5. Discussion of Results

Overall, the experiments demonstrate that increasingly sophisticated retrieval strategies produced progressively stronger results, although the magnitude of improvement varied between techniques.

The main findings can be summarised as follows:

- **TF-IDF substantially outperformed TF**, confirming the importance of inverse document frequency weighting.
- **TF-IDF tuning produced limited additional improvement**, indicating that the baseline configuration was already strong.
- **Lemur TF-IDF did not outperform standard TF-IDF**, even after parameter tuning.
- **Field weighting provided a modest improvement**, with approximately 26% title weighting and 74% text weighting giving the strongest MAP.
- **RM3 produced the largest improvement**, increasing MAP from 0.318897 to 0.353066.
- **RM3 achieved the highest MAP, P@5, P@10, and nDCG@10** among the tested configurations.
- The best RM3 configuration was **30 feedback documents and 15 feedback terms**.
- RM3 required substantially more search time, but the improvement in retrieval effectiveness justified the additional computational cost for the final system.

Based on these findings, **RM3 with 30 feedback documents and 15 feedback terms was selected as the final retrieval configuration**. This configuration provided the strongest overall balance of retrieval effectiveness across the principal evaluation measures and was therefore used to generate the final submission results.

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